



Chemistry Paper

- The grid given below represents part of the periodic table. Study it and answer the questions that follow. The letters do not represent the actual symbols of elements.

M					N	P	T	
R								

- Select a letter which represents an element that loses electrons most readily.
- Give a reason for your answer. (3 marks)
- Explain why the atomic radius of P is found to be smaller than that of N. (2 marks)
- Element M reacts with water at room temperature to produce 100 cm³ of gas. Determine the mass of M which was reacted with water. (Molar gas volume at room temperature is 100 cm³. Relative atomic mass of M is 30. (4 marks)
- Use the information in the table below to answer the questions that follow. The letters are not the symbols of the elements.

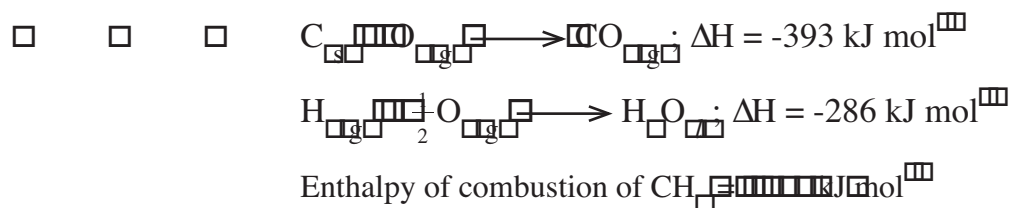
Element	State of oxide at room temperature	Type of oxide	Bonding in oxide
U	Solid	Acidic	Covalent
W	Solid	Basic	Ionic
X	Liquid	Neutral	Covalent
Y	Gas	Neutral	Covalent

- Identify a letter which represents an element in the table that could be calcium. (1 mark)
- carbon or sulphur. Give a reason in each case. (2 marks)
- Calcium. (1 mark)
- Reason. (1 mark)
- Carbon. (1 mark)
- Reason. (1 mark)
- Sulphur. (1 mark)
- Reason. (1 mark)



☐ (a) (i) What is meant by the term 'Enthalpy of formation'? (1 mark)

☐ ☐ ☐ ☐ The enthalpies of combustion of carbon, methane and hydrogen are indicated below ☐



☐ ☐ ☐ I Draw an energy cycle diagram that links the enthalpy of formation of methane to the enthalpies of combustion of carbon, hydrogen and methane. (3 marks)

II Determine the enthalpy of formation of methane. (2 marks)

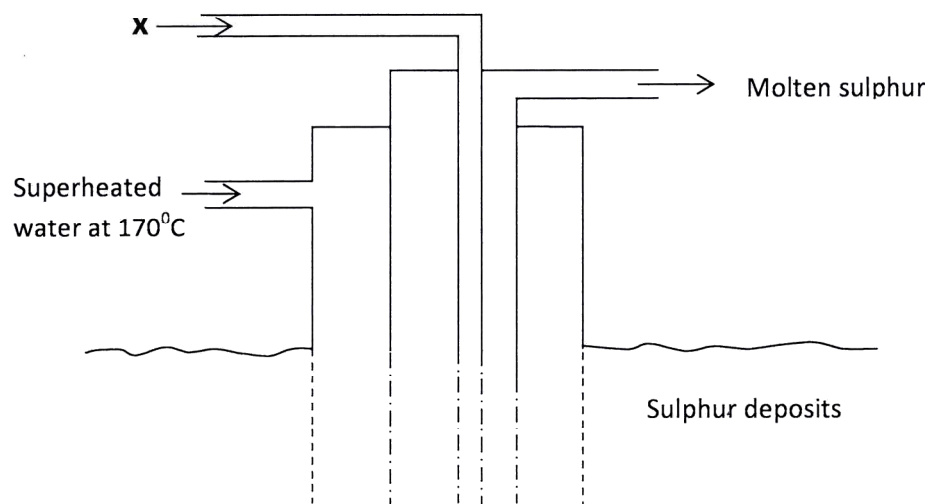
☐ ☐ ☐ An experiment was carried out where different volumes of dilute hydrochloric acid and aqueous sodium hydroxide both at 25°C were mixed and stirred with a thermometer. The highest temperature reached by each mixture was recorded in the table below.

Volume of hydrochloric acid / cm^3									
Volume of sodium hydroxide / cm^3									
Highest temperature of mixture ($^\circ\text{C}$)									



- ☐ ☐ ☐ The molar enthalpy of neutralisation between hydrochloric acid and ammonia solution ☐
☐ ☐ was found to be $-46.0 \text{ kJ mol}^{-1}$ while that of hydrochloric acid and sodium hydroxide ☐
☐ ☐ was $-57.1 \text{ kJ mol}^{-1}$. Explain the difference in these values ☐ ☐ (3 marks)

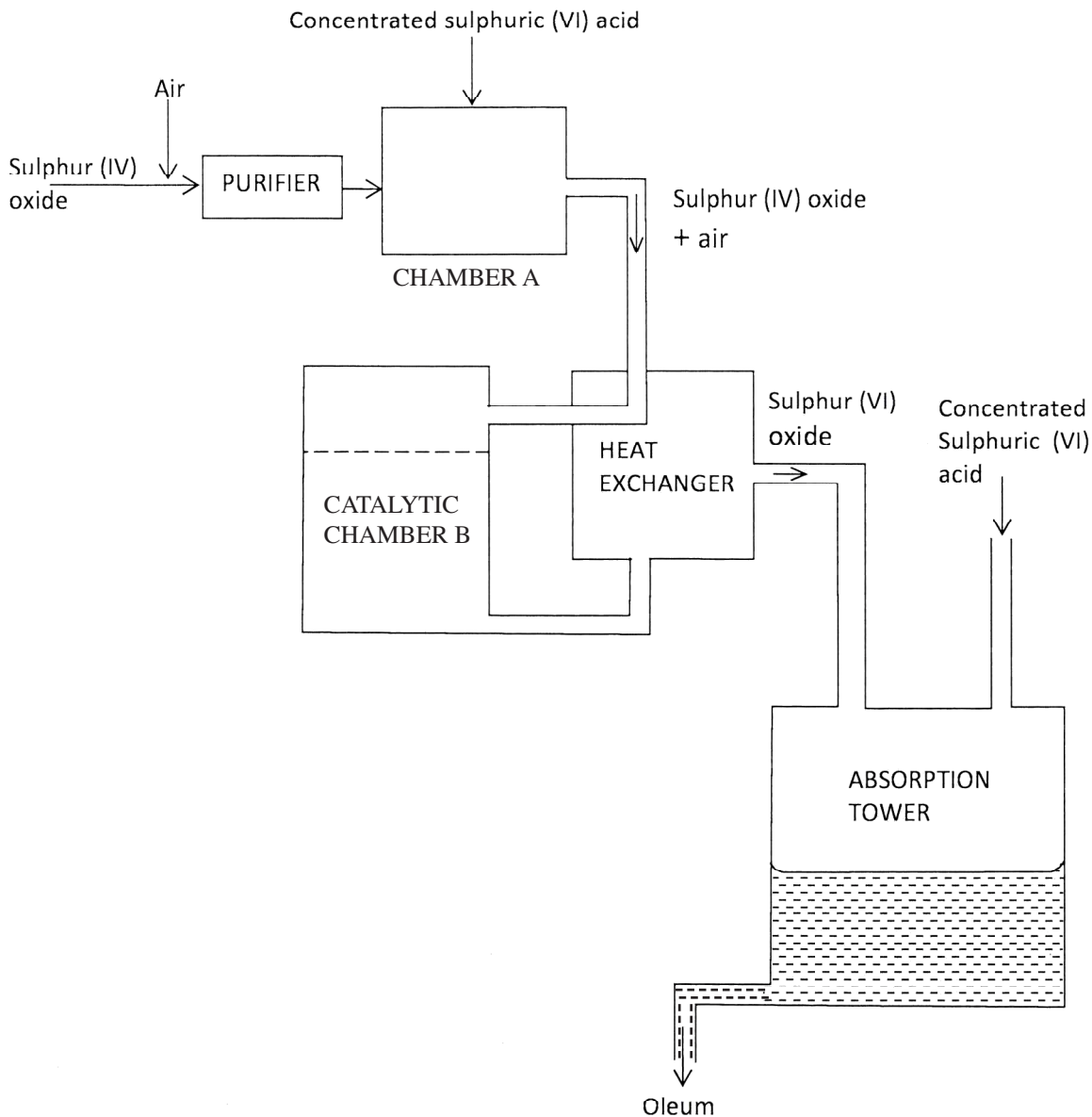
- ☐ ☐ ☐ The diagram below shows the Frasch process used for extraction of sulphur ☐ ☐
☐ ☐ Use it to answer the questions that follow ☐



- ☐ ☐ ☐ Identify **x** ☐ ☐ (1 mark)
- ☐ (ii) Why is it necessary to use super heated water in this process? (1 mark)
- ☐ ☐ ☐ State two physical properties of sulphur that makes it possible for it to be ☐
☐ ☐ extracted by this method ☐ ☐ (2 marks)



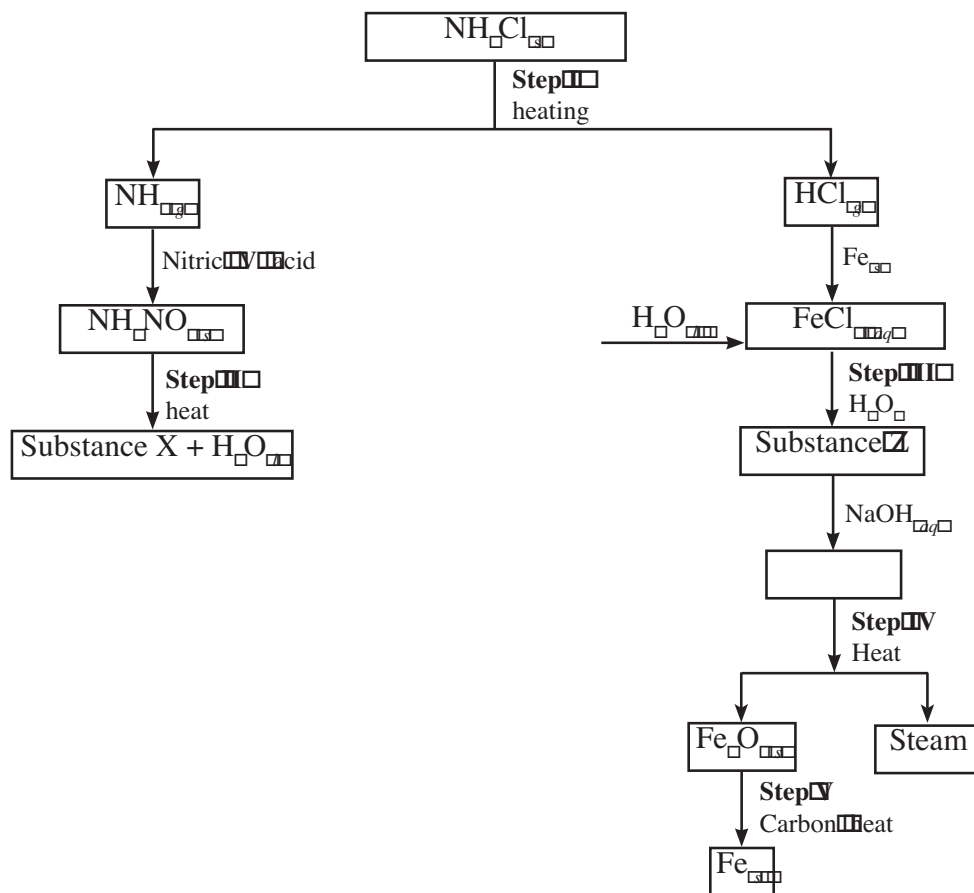
- ☐ ☐ ☐ ☐ The diagram below shows part of the processes in the manufacture of sulphuric (VI) acid. Study it and answer the questions that follow.



- ☐ ☐ ☐ ☐ Write an equation for the formation of sulphur (IV) oxide from sulphur. (1 mark)
- (ii) What is the role of concentrated sulphuric (VI) acid in chamber A? (1 mark)
- ☐ ☐ ☐ ☐ Name two catalysts that can be used in the catalytic chamber B. (2 marks)
- ☐ ☐ ☐ ☐ State two roles of the heat exchanger. (2 marks)
- ☐ ☐ ☐ ☐ Explain one way in which sulphur (IV) oxide is a pollutant. (1 mark)



- (a) Describe **one** method that can be used to distinguish between sodium sulphate and sodium hydrogen sulphate. (3 marks)
- (b) Describe how a pure sample of lead (II) sulphate can be prepared in the laboratory starting with lead metal. (4 marks)
- (c) Study the flow chart below and answer the questions that follow:



- Write an equation for the reaction in step I. (1 mark)
- II. step IV. (1 mark)
- State the observation made in step III and explain. (3 marks)
- Name another substance that can be used in step VIII. (1 mark)
- (a) Distinguish between a neutron and a proton. (1 mark)
- (b) What is meant by a radioactive substance? (1 mark)
- State two dangers associated with radioactive substances in the environment. (2 marks)

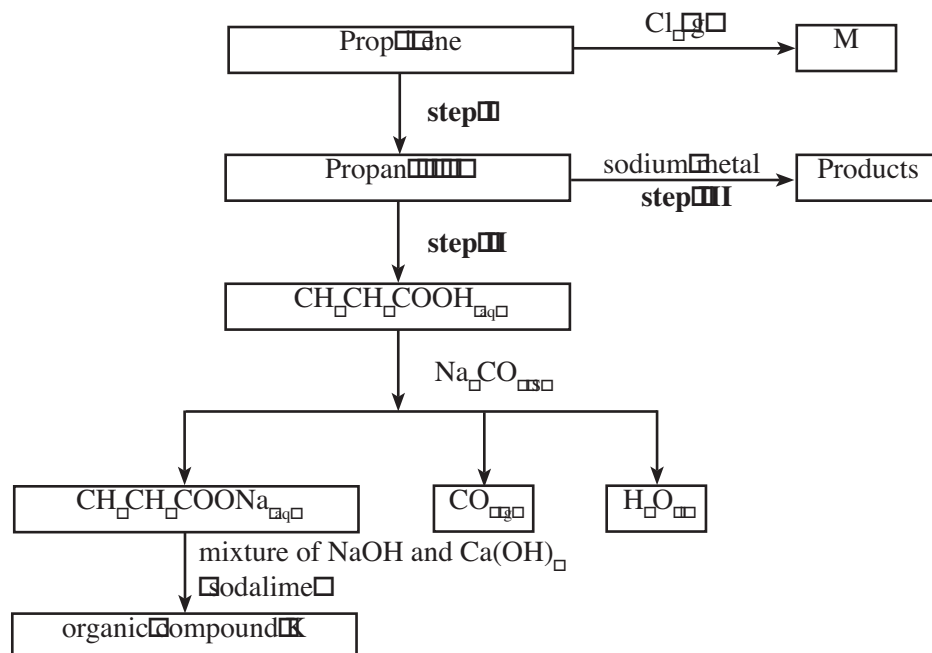


- ☐ ☐ ☐ The two isotopes of hydrogen, deuterium (${}^2_1\text{D}$) and tritium (${}^3_1\text{T}$) react to form element Y and neutron particles according to the equation below.
☐ ☐ ${}^2_1\text{D} + {}^3_1\text{T} \rightarrow {}^a_b\text{Y} + {}^1_0\text{n}$
- ☐ ☐ ☐ ☐ What is the atomic mass of Y? ☐ ☐ (1 mark)
- ☐ ☐ ☐ ☐ ☐ number of Y? ☐ ☐ (1 mark)
- ☐ ☐ ☐ ☐ What name is given to the type of reaction undergone by the isotopes of hydrogen? (1 mark)
- (e) (i) What is meant by half-life of a radioactive substance? (1 mark)
 (ii) 288g of a radioactive substance decayed to 9g in 40 days. Determine the half-life of the radioactive substance. ☐ ☐ (2 marks)

☐ ☐ ☐ Give the systematic names for the following compounds.

- (i) $\text{CH}_3\text{CH}_2\text{COOH}$; (1 mark)
- (ii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHCH}_3$ ☐ ☐ (1 mark)
- (iii) $\text{CH}_3\text{C}(\text{CH}_3)_2\text{CH}_2\text{CH}_3$ ☐ ☐ (1 mark)

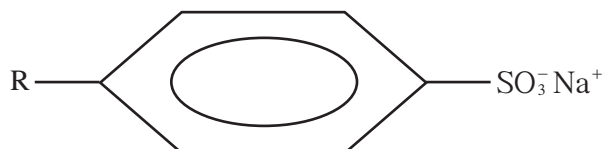
(b) Study the flow chart below and use it to answer the questions that follow:



- ☐ ☐ ☐ ☐ Identify the organic compound K. ☐ ☐ (1 mark)



- ☐ ☐ ☐ Write the formula of Mn^{2+} ☐ ☐ mark
- ☐ ☐ ☐ Give one reagent that can be used in ☐ ☐ mark
- ☐ ☐ ☐ step ☐ ☐ ☐ mark
- ☐ ☐ ☐ step ☐ ☐ ☐ mark
- ☐ ☐ ☐ Write the equation of the reaction in step ☐ ☐ ☐ mark
- ☐ ☐ ☐ The structure below represents a type of a cleansing agent ☐ ☐ mark



Describe how the cleansing agent removes grease from a piece of cloth. (3 marks)